

AI ASSISTANT CODING

ASSIGNMENT-14.2

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Enrollment no: 2303A52103

Batch: 33

Semester: VI

Branch: CSE(AIML)

Lab 14: Web Design Application – AI-Assisted HTML/CSS/JS Generation

Lab Objectives

- Design functional, visually appealing web applications using HTML, CSS, and JavaScript with AI assistance.
- Apply responsive, accessible, and interactive design principles.
- Create practical UI components for real-world web applications.
- Use AI to optimize layout, UX, and performance.

Task Description 1 – (Landing Page Design)

- Create a simple landing page for a product or service. Include a header, hero section, and footer.

Requirements:

- Use HTML and CSS to structure the page.
- Use AI to suggest layout alignment and typography.
- Ensure the page is responsive across devices.

```

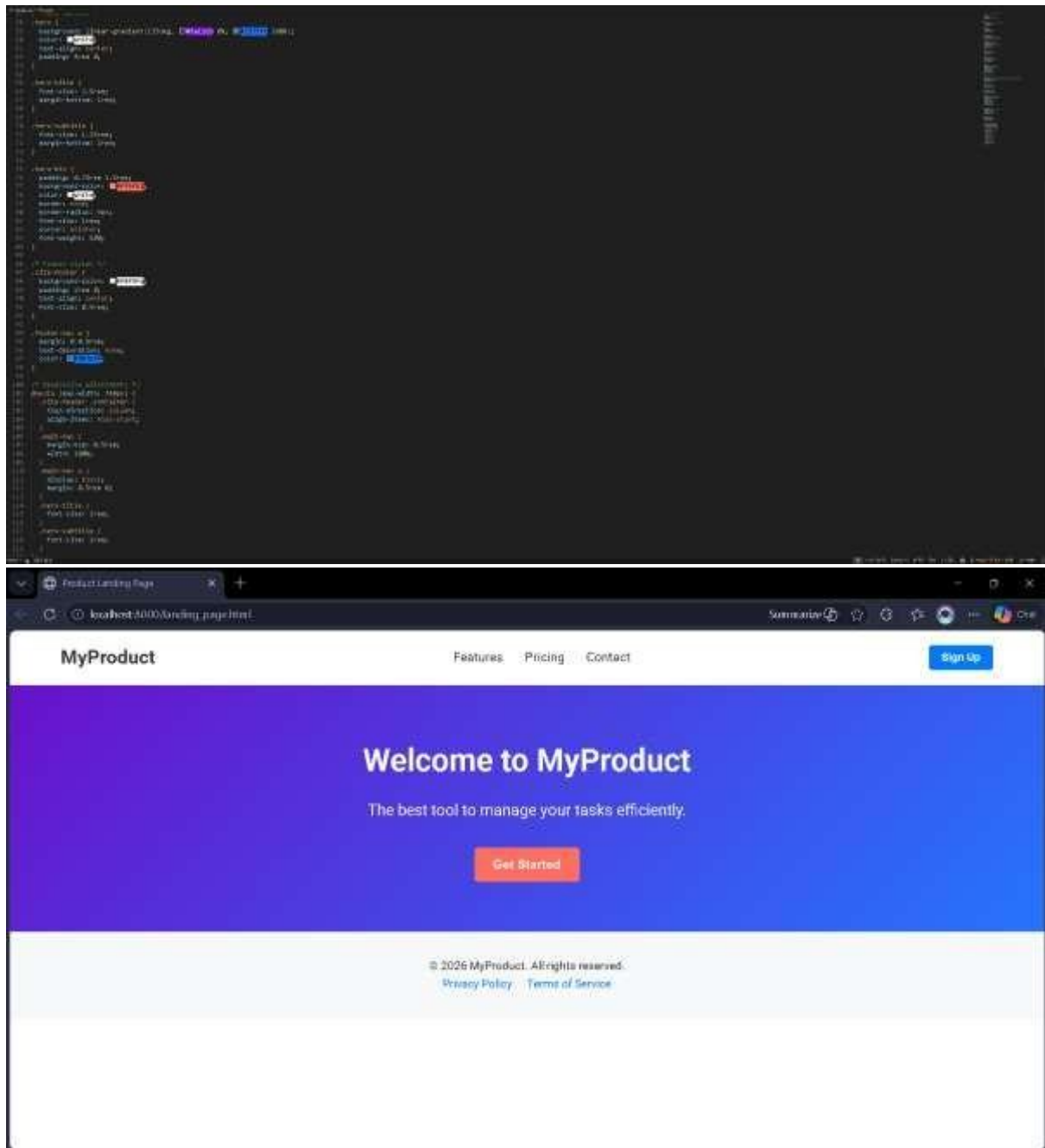
1 // Importing React
2 import React from 'react';
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4 // Importing ReactDOM
5 import ReactDOM from 'react-dom';
6
7 // Importing PropTypes
8 import PropTypes from 'prop-types';
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10 // Importing classNames
11 import classNames from 'classnames';
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13 // Importing lodash
14 import _ from 'lodash';
15
16 // Importing axios
17 import axios from 'axios';
18
19 // Importing moment
20 import moment from 'moment';
21
22 // Importing uuid
23 import uuid from 'uuid';
24
25 // Importing bcrypt
26 import bcrypt from 'bcrypt';
27
28 // Importing jwt
29 import jwt from 'jsonwebtoken';
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31 // Importing config
32 import config from './config';
33
34 // Importing constants
35 import constants from './constants';
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101 import errorHandler from './errorHandler';

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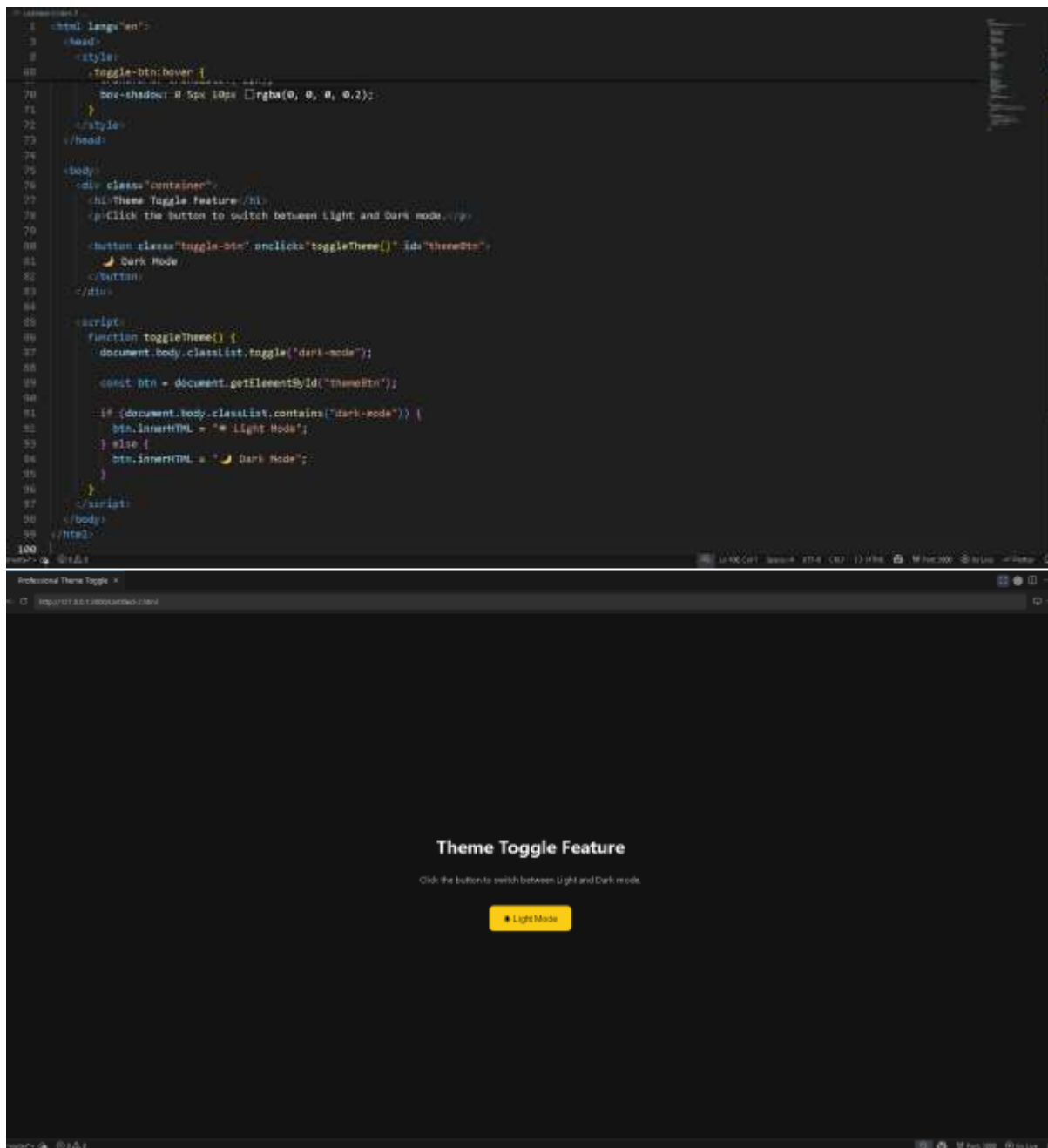
Task Description 2 – (Theme Toggle Feature)

- Provide a button to switch between light and dark themes..

Requirements:

- Use JavaScript to toggle CSS classes.
- Use AI assistance to manage color contrast and accessibility.
- Ensure theme preference applies instantly on click.

```
1 <!doctype html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8" />
5 <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6 <title>Professional Theme Toggle</title>
7
8 <style>
9 /* CSS Variables for Themes */
10 :root {
11   --bg-color: #ffffff;
12   --text-color: #002222;
13   --button-bg: #25838b;
14   --button-text: #ffffff;
15 }
16
17 body.dark-mode {
18   --bg-color: #121212;
19   --text-color: #f5f5f5;
20   --button-bg: #f0c000;
21   --button-text: #000000;
22 }
23
24 /* Page Styling */
25 body {
26   margin: 0;
27   font-family:
28     Segoe UI,
29     Arial,
30     sans-serif;
31   background: var(--bg-color);
32   color: var(--text-color);
33   transition:
34     background 0.4s,
35     color 0.4s;
36 }
37
38 /* Container */
39 .container {
40   display: flex;
41   flex-direction: column;
42   justify-content: center;
43   align-items: center;
44   height: 100vh;
45   text-align: center;
46 }
47
48 /* Heading */
49 h1 {
50   margin-bottom: 10px;
51 }
52
53 /* Toggle Button */
54 .toggle-btn {
55   margin-top: 20px;
56   padding: 10px 25px;
57   font-size: 16px;
58   border: none;
59   border-radius: 5px;
60   background: var(--button-bg);
61   color: var(--button-text);
62   cursor: pointer;
63   transition:
64     transform 0.2s,
65     box-shadow 0.2s;
66 }
67
68 .toggle-btn:hover {
69   transform: translateY(-2px);
70 }
```

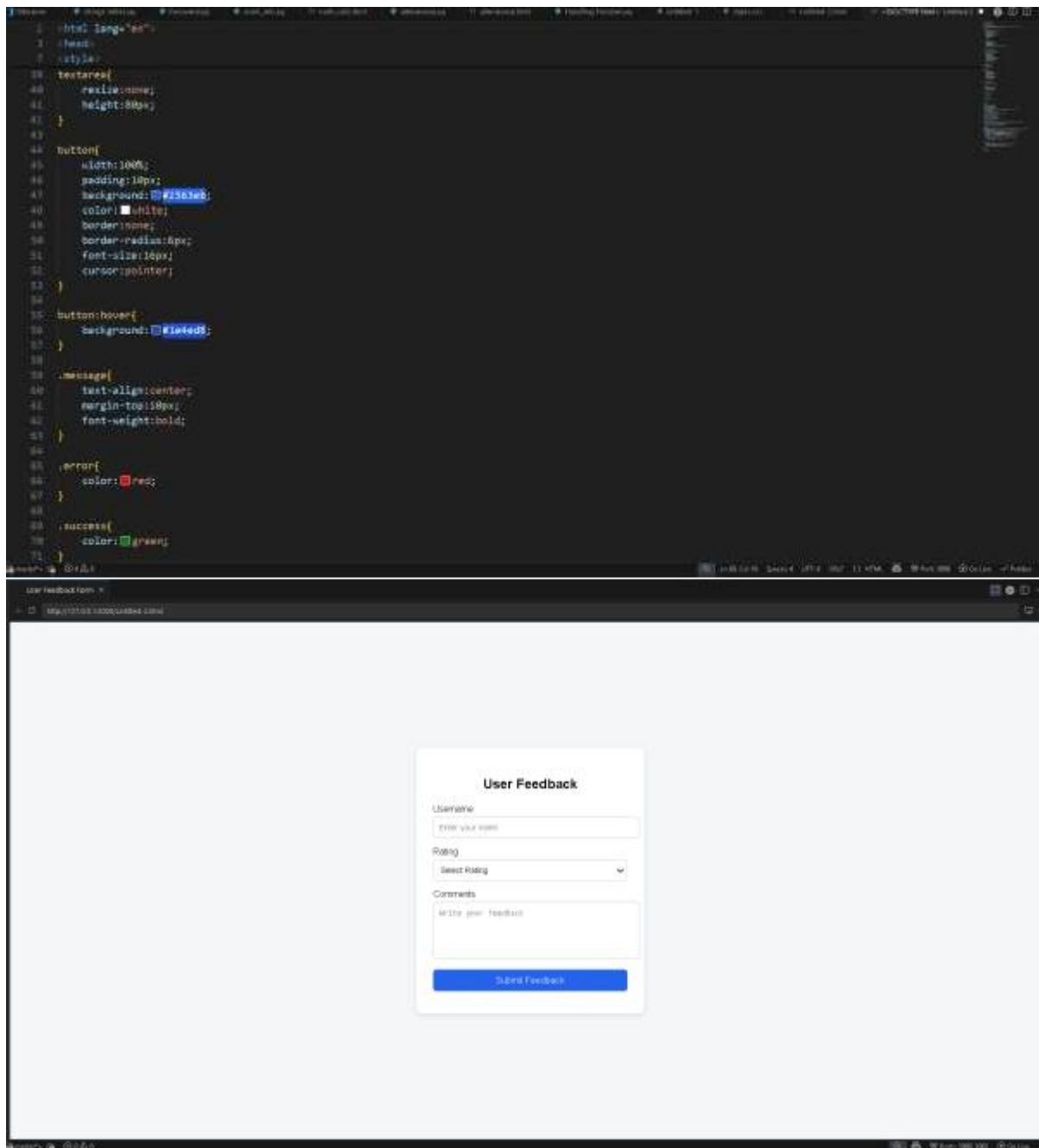


Task Description 3 – (User Feedback Form)

- Create a feedback form with fields: Username, Rating, Comments.

Requirements:

- Use JavaScript to validate required fields.
- Display success or error messages dynamically.
- Use AI to improve form usability and layout.



Task Description - 4 (Image Gallery with JavaScript)

- Task: Display a small image gallery on a web page.

Requirements:

- Use JavaScript to dynamically load images.
- Add next/previous navigation buttons.
- Use AI to optimize layout responsiveness.

```

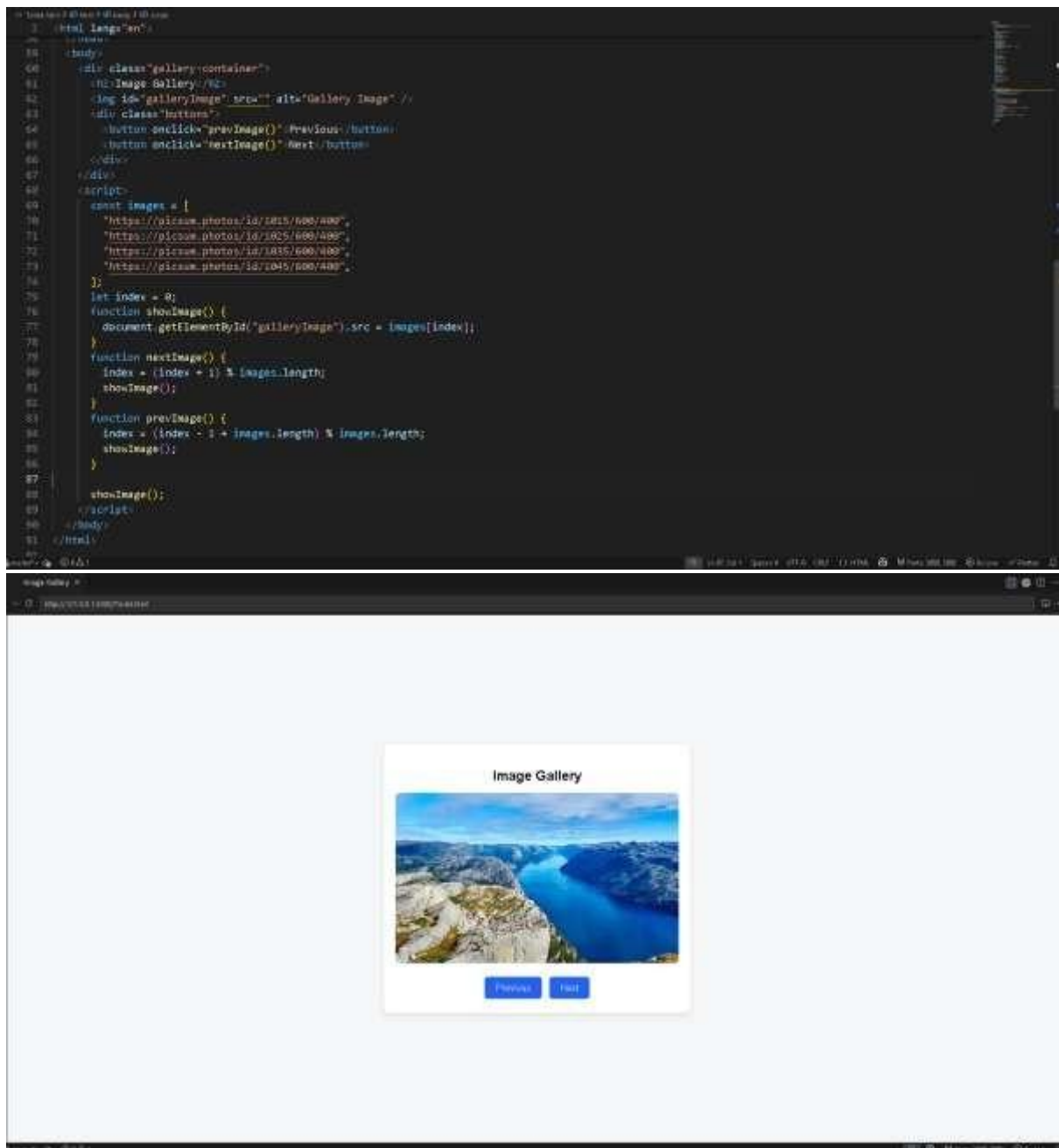
1 <!doctype html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8" />
5 <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6 <title>Image Gallery</title>
7 <style>
8 <body>
9 font-family: Arial, sans-serif;
10 background: #f4f6f8;
11 margin: 0;
12 display: flex;
13 justify-content: center;
14 align-items: center;
15 height: 100vh;
16 </body>
17 <div class="gallery-container">
18 text-align: center;
19 background: white;
20 padding: 10px;
21 border-radius: 10px;
22 box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);
23 max-width: 500px;
24 width: 90%;
25 </div>
26 <h2>
27 margin-bottom: 15px;
28 </h2>
29 <img>
30 width: 100%;
31 height: 200px;
32 object-fit: cover;
33 border-radius: 5px;
34 </img>
35 <div class="buttons">

```

```

36 <div class="buttons">
37 margin-top: 15px;
38 </div>
39 <div class="button">
40 padding: 10px 20px;
41 margin: 5px;
42 border: none;
43 border-radius: 5px;
44 background: #f5f5f5;
45 color: white;
46 font-size: 16px;
47 cursor: pointer;
48 </div>
49 <div class="button-hover">
50 background: #e0e0e0;
51 </div>
52 <div class="media (max-width: 500px)">
53 <img>
54 height: 200px;
55 </img>
56 </div>
57 </style>
58 </head>
59 <body>
60 <div class="gallery-container">
61 <h2>Image Gallery</h2>
62 <img id="galleryImage" src="" alt="Gallery image" />
63 <div class="buttons">
64 <button onclick="prevImage()">Previous</button>
65 <button onclick="nextImage()">Next</button>
66 </div>
67 </div>
68 </body>
69 </html>

```

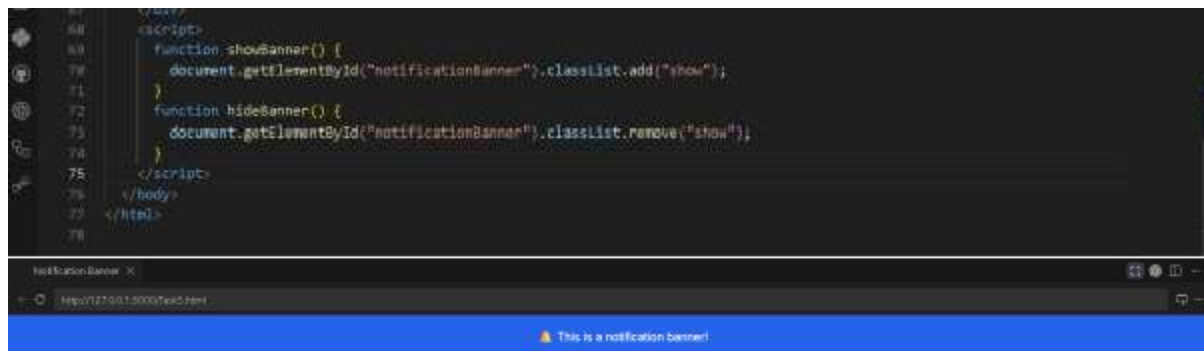
Task Description - 5: (Notification Banner)

- Display a notification banner at the top of the page.

Requirements:

- Use JavaScript to show and hide the banner.
- Style the banner using CSS.
- Use AI to ensure smooth transitions and clean UI behavior.

```
1 <!doctype html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8" />
5 <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6 <title>Notification Banner</title>
7 <style>
8 /* Page styling */
9 body {
10 margin: 0;
11 font-family: Arial, sans-serif;
12 }
13 /* Notification banner */
14 .banner {
15 position: fixed;
16 top: -100px;
17 left: 0;
18 width: 100%;
19 background: #156580;
20 color: white;
21 text-align: center;
22 padding: 10px;
23 font-size: 14px;
24 transition: top 0.5s ease;
25 }
26 /* when banner is visible */
27 .banner.show {
28 top: 0;
29 }
30 /* Close button */
31 .close-btn {
32 position: absolute;
33 right: 10px;
34 top: 10px;
35 cursor: pointer;
36 }
37 </style>
38 </head>
39 <body>
40 <div id="notificationBanner" class="banner">
41 <span class="close-btn" onclick="hideBanner()" />
42 </div>
43 <div class="container">
44 <h1>Notification Banner Example</h1>
45 <p>Click the button to display the notification.</p>
46 <button onclick="showBanner()">Show Notification</button>
47 </div>
48 </body>
49 </html>
50 </script>
```



Notification Banner Example

Click the button to display the notification.

Show Notification